Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · www.solaress.org

ELECTRICAL WORKS · INSPECTION & TEST

General Electrical Works Report

Report no.

Complete one report for every electrical job. Record what was installed, test every circuit before energising, and ask the customer to sign before the team leaves. Write "N/A" where an item does not apply.

A Site & Job Details

Customer / site name

Job type
☐ New installation ☐ Modification ☐ Repair ☐ Inspection

Site address

Customer contact (name, phone)

Supply system
☐ 1-phase 230 V ☐ 3-phase 400 V

PEA meter size / meter no.

Main breaker rating (A) / board name

Contract / PO reference

Scope of work (short description)

1 Service Team

Officer in charge | **Position** | **Phone** | **Number of staff on site**

#	Team member	Role	Qualification / licence no.	Signature
1				
2				
3				
4				
5				

2 Operation Time

Date	Arrival	Work start	Work finish	Total duration (h)

3 Safety & Isolation

TICK BEFORE WORK STARTS

- ☐ Site permit / customer approval obtained
- ☐ Circuits isolated at the supply breaker
- ☐ Absence of voltage verified with a tested meter
- ☐ Customer informed of power interruption time
- ☐ PPE worn: helmet, safety glasses, insulated gloves, safety shoes
- ☐ Lock-out / tag-out applied to every isolated device
- ☐ Insulated tools used; work area barricaded
- ☐ Power restored and loads checked after work

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4 Specification of Equipment Installed

4.1 Circuit breakers (MCCB / MCB / RCBO)

#	Board / location	Circuit / load	Type	Brand / model	Rating (A)	Poles (1P–4P)	Breaking capacity (kA)	RCD (mA)
1			☐ MCCB ☐ MCB ☐ RCBO					
2			☐ MCCB ☐ MCB ☐ RCBO					
3			☐ MCCB ☐ MCB ☐ RCBO					
4			☐ MCCB ☐ MCB ☐ RCBO					
5			☐ MCCB ☐ MCB ☐ RCBO					
6			☐ MCCB ☐ MCB ☐ RCBO					
7			☐ MCCB ☐ MCB ☐ RCBO					
8			☐ MCCB ☐ MCB ☐ RCBO					
9			☐ MCCB ☐ MCB ☐ RCBO					
10			☐ MCCB ☐ MCB ☐ RCBO					
11			☐ MCCB ☐ MCB ☐ RCBO					
12			☐ MCCB ☐ MCB ☐ RCBO					
13			☐ MCCB ☐ MCB ☐ RCBO					
14			☐ MCCB ☐ MCB ☐ RCBO					

4.2 Cables

#	Circuit (from → to)	Cable type (THW, NYY, VCT, CV)	Size (mm²) × cores	Conductor	Colour code	Length (m)	Remarks
1				☐ Cu ☐ Al			
2				☐ Cu ☐ Al			
3				☐ Cu ☐ Al			
4				☐ Cu ☐ Al			
5				☐ Cu ☐ Al			
6				☐ Cu ☐ Al			
7				☐ Cu ☐ Al			
8				☐ Cu ☐ Al			
9				☐ Cu ☐ Al			
10				☐ Cu ☐ Al			
11				☐ Cu ☐ Al			
12				☐ Cu ☐ Al			
13				☐ Cu ☐ Al			
14				☐ Cu ☐ Al			

Colour code (TIS 11 / EIT standard): L1 brown · L2 black · L3 grey · N blue · PE green-yellow. Write the colours used if different.

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4.3 Wiring method

#	Circuit / route	Air	PVC	IMC	EMT	Tray	Conduit size	Supports, spacing & remarks
1		☐	☐	☐	☐	☐		
2		☐	☐	☐	☐	☐		
3		☐	☐	☐	☐	☐		
4		☐	☐	☐	☐	☐		
5		☐	☐	☐	☐	☐		
6		☐	☐	☐	☐	☐		
7		☐	☐	☐	☐	☐		
8		☐	☐	☐	☐	☐		
9		☐	☐	☐	☐	☐		
10		☐	☐	☐	☐	☐		

Air = open wiring on clips or insulators. Outdoor and exposed runs: IMC; indoor dry areas: EMT or PVC. Fill each conduit to no more than 40%.

4.4 Terminals & connections

#	Location / device	Terminal type	Torque spec (N·m)	Torqued	Insulated	Remarks
1		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
2		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
3		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
4		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
5		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
6		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
7		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
8		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
9		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	
10		☐ Lug ☐ Ferrule ☐ Busbar ☐ Other		☐	☐	

Tighten to the manufacturer's torque and mark each terminal after torquing. Use crimped lugs or ferrules on every stranded conductor; no bare strands.

5 Photo Record

SUBMITTED AS A SEPARATE FILE

Send the photos as a separate file with this report, with the date and time stamp on. Name the file: **Report no._Site name_Photos_YYYYMMDD**.

Photo file name / link _____

Number of photos _____

Sent to / date sent _____

Photos included in the file (tick)

- ☐ 1. Board / MDB before work
- ☐ 2. Board / MDB after work, door open
- ☐ 3. Breaker labels and ratings
- ☐ 4. Cable runs and conduit
- ☐ 5. Terminals and connections, close-up
- ☐ 6. Meter readings during testing
- ☐ 7. Nameplates of new equipment
- ☐ 8. Other: _____

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6 Testing

6.1 Insulation resistance

#	Circuit	Test voltage (V DC)	L-N (MΩ)	L-PE (MΩ)	N-PE (MΩ)	L-L (MΩ)	Minimum	Result
1							≥ 1 MΩ	☐ Pass ☐ Fail
2							≥ 1 MΩ	☐ Pass ☐ Fail
3							≥ 1 MΩ	☐ Pass ☐ Fail
4							≥ 1 MΩ	☐ Pass ☐ Fail
5							≥ 1 MΩ	☐ Pass ☐ Fail
6							≥ 1 MΩ	☐ Pass ☐ Fail
7							≥ 1 MΩ	☐ Pass ☐ Fail
8							≥ 1 MΩ	☐ Pass ☐ Fail
9							≥ 1 MΩ	☐ Pass ☐ Fail
10							≥ 1 MΩ	☐ Pass ☐ Fail

Test at 500 V DC for circuits up to 500 V (IEC 60364-6), with loads and electronic devices disconnected. Minimum 1 MΩ per circuit.

6.2 Voltage and current at connection points

#	Connection point / board	V L1-N	V L2-N	V L3-N	V L-L	I L1 (A)	I L2 (A)	I L3 (A)	I N (A)
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									

Measure at each new or modified connection point under normal load. Voltage limits: 230 V ±10% (L-N), 400 V ±10% (L-L).

6.3 Earthing and protection

Test	Measured	Target	Result
Earth electrode resistance (Ω)		≤ 5 Ω	☐ Pass ☐ Fail
RCD trip test at rated current (ms)		≤ 300 ms	☐ Pass ☐ Fail
RCD test button trips		Trips	☐ Pass ☐ Fail
Polarity correct at all outlets / loads		Correct	☐ Pass ☐ Fail
Phase rotation (3-phase)		L1-L2-L3	☐ Pass ☐ Fail
Neutral–earth voltage at load (V)		< 2 V	☐ Pass ☐ Fail

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7 Other Reports Attached

- ☐ Thermal scan report
- ☐ As-built single-line diagram
- ☐ Load schedule / panel schedule
- ☐ Material list / delivery notes
- ☐ Equipment datasheets / certificates
- ☐ PEA or authority documents
- ☐ Commissioning checklist
- ☐ Other: _____

Notes on attached reports

8 Findings & Recommendations

#	Finding	Recommendation	Priority	Est. cost (THB, excl. VAT)
1			☐ H ☐ M ☐ L	
2			☐ H ☐ M ☐ L	
3			☐ H ☐ M ☐ L	
4			☐ H ☐ M ☐ L	
5			☐ H ☐ M ☐ L	
6			☐ H ☐ M ☐ L	
7			☐ H ☐ M ☐ L	
8			☐ H ☐ M ☐ L	
9			☐ H ☐ M ☐ L	

Warranty on workmanship

Next inspection due

9 Customer Comments

10 Sign-off

Prepared by · Solar ESS

Name | Position

Date | Signature

Acknowledged by · Customer

Name | Position

Date | Signature

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